

Impedance-Matching vs. Robin-Robin Coupling Conditions in Schwarz Methods

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1 Setting

Consider a solid body $\Omega \subset \mathbb{R}^3$ decomposed into two subdomains Ω_1 and Ω_2 that are integrated independently during a Schwarz iteration. Two geometric configurations are used in Norma.jl:

Non-overlap. The subdomains partition Ω and share an interface $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ on which coupling is imposed. Let $\mathbf{n}_k$ be the outward unit normal to Ω_k on Γ (so $\mathbf{n}_1 = -\mathbf{n}_2$), and let

$$\mathbf{t}_k = \boldsymbol{\sigma}_k \mathbf{n}_k$$

denote the traction on Ω_k 's side of the interface. Continuity of the coupled solution requires

$$\mathbf{u}_1 = \mathbf{u}_2 \quad (\text{kinematic continuity}), \quad (1)$$

$$\mathbf{t}_1 + \mathbf{t}_2 = \mathbf{0} \quad (\text{traction balance}). \quad (2)$$

Classical Dirichlet-Neumann (DN) Schwarz picks up (1) on one side and (2) on the other. The *Robin-Robin* (RR) and *impedance* variants each replace both Dirichlet and Neumann conditions with a linear combination of displacement and traction, which offers extra tuning and, in the impedance case, wave absorption.

Overlap. The subdomains cover parts of Ω that properly intersect: $\Omega_1 \cap \Omega_2 \neq \emptyset$. Each Ω_k has a portion of its boundary, $\Gamma_k \subset \Omega_j$ with $j \neq k$, that lies strictly in the interior of the partner. That boundary is not shared with Ω_j , so no mutual traction is available there; on the other hand, the partner provides a well-defined interior state at every node of Γ_k . Classical overlap Schwarz enforces kinematic matching

$$\mathbf{u}_k = \mathbf{u}_j \quad \text{on } \Gamma_k,$$

either pointwise (strong interpolation) or in an L^2 sense (weak projection). The *impedance overlap* variant instead leaves the overlap degrees of freedom free and drives $\mathbf{u}_k$ towards $\mathbf{u}_j$ through an impedance-matching penalty, as detailed in Section 4.

2 Robin-Robin (RR) Non-Overlap Schwarz

2.1 Continuous formulation

On each side of the interface, the user specifies a Robin parameter $\alpha > 0$ and imposes

$$\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma, \quad k = 1, 2, \quad (3)$$

with the forcing $\mathbf{g}_k$ set from the partner subdomain's state at the previous Schwarz iterate:

$$\mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + \alpha \mathbf{u}_j^{(n)} \quad j \neq k. \quad (4)$$

When the Schwarz iteration converges, $\mathbf{u}_k = \mathbf{u}_j$ and $\mathbf{t}_k = -\mathbf{t}_j$ so that (3) reproduces (1) and (2) simultaneously. The combined “elastic-spring” interface smooths out the DN mismatch that drives DN Schwarz towards a fixed point.

2.2 Discrete implementation

Let $\mathbf{W} = \int_{\Gamma} \mathbf{N} \mathbf{N}^{\top} d\Gamma$ be the surface mass (square-projector) matrix and let $\mathbf{P}$ be the Dirichlet projector mapping the partner's interface trace to this subdomain's interface nodes. The RR boundary condition contributes to the solid subproblem on Ω_k as follows:

Self stiffness term. From (3), the quantity $\alpha \mathbf{u}_k$ acts as an elastic boundary stiffness which is assembled into the global tangent:

$$\mathbf{K}_{rs} = \alpha \mathbf{W} \quad (\text{diagonal of } \Gamma). \quad (5)$$

Partner contribution (right-hand side). The forcing $\mathbf{g}_k$ from (4) is accumulated into the boundary-force vector of Ω_k :

$$\mathbf{f}_{\Gamma,k} \leftarrow \mathbf{f}_{\Gamma,k} - \mathbf{t}_j^{(n)}|_{\Gamma} + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j^{(n)}. \quad (6)$$

In Norma.jl the partner traction is negated once by `get_dst_force` and the self term $\alpha \mathbf{W} \mathbf{u}_k$ is added back via `apply_robin_bcs_internal_force!`.

2.3 Parameter tuning

α has units of stiffness per unit area ($\text{Pa/m} = \text{N/m}^3$). It trades off traction and displacement error:

- $\alpha \rightarrow \infty$: the Robin condition approaches a pure Dirichlet constraint; no traction feedback from the partner.
- $\alpha \rightarrow 0$: the Robin condition approaches a pure Neumann (traction) constraint; no displacement feedback.
- An optimal α minimizes the Schwarz convergence rate. Closed-form expressions have been studied in the optimized-Schwarz literature but depend on details of the discretization and problem. In practice α is tuned empirically; the scaling $\alpha \sim E/h$ (Young's modulus over a representative element size) has been effective on the problems exercised here.

2.4 Limitations for dynamic problems

For dynamic (wave-propagation) problems, (3) behaves as a purely elastic interface. An incoming wave from Ω_j is partly reflected back into Ω_j and partly transmitted into Ω_k , with reflection coefficient determined by the interface impedance mismatch. Because α has no physical correspondence to the material's wave impedance, reflections in general are non-zero and can drive energy growth when the two subdomains use different time integrators (e.g. an implicit/explicit pairing across the interface).

3 Impedance Non-Overlap Schwarz

3.1 Motivation: non-reflecting boundary in 1D

Consider a semi-infinite elastic bar with density ρ and P-wave modulus $M = \lambda + 2\mu$, supporting a right-going traveling wave $u(x, t) = f(x - c_p t)$ with $c_p = \sqrt{M/\rho}$. Differentiating in x and t gives the kinematic identity $\dot{u} = u_t = -c_p u_x$. Combined with 1D Hooke's law $\sigma = M u_x$ this yields

$$\sigma = M u_x = -\frac{M}{c_p} \dot{u} = -\rho c_p \dot{u}. \quad (7)$$

Define the *characteristic acoustic impedance*

$$Z = \rho c_p = \sqrt{\rho(\lambda + 2\mu)}. \quad (8)$$

Then for a purely outgoing (right-going) wave,

$$\sigma + Z \dot{u} = 0, \quad (9)$$

which is the Lysmer-Kuhlemeyer non-reflecting boundary condition. Imposing (9) on the interface prevents spurious reflections of outgoing waves and is the key ingredient in the impedance variant.

3.2 Continuous formulation

Norma.jl's impedance Schwarz BC augments the Robin condition with the impedance term from (9):

$$\mathbf{t}_k + Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma. \quad (10)$$

The forcing is built from the partner state:

$$\mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + Z \dot{\mathbf{u}}_j^{(n)} + \alpha \mathbf{u}_j^{(n)}. \quad (11)$$

At convergence, $\mathbf{u}_k = \mathbf{u}_j$, $\dot{\mathbf{u}}_k = \dot{\mathbf{u}}_j$, and $\mathbf{t}_k = -\mathbf{t}_j$, so the impedance and displacement-penalty terms cancel and (10) again reproduces continuity and traction balance.

Setting $\alpha = 0$ yields the pure impedance condition; the displacement-penalty term $\alpha \mathbf{u}$ is retained as an option to regularize the quasi-static limit, where $\dot{\mathbf{u}} \rightarrow \mathbf{0}$ and (10) would otherwise reduce to the Neumann condition only.

3.3 Discrete implementation

For a Newmark integrator with parameters β , γ and time step Δt , the velocity is a linear function of displacement at the new step,

$$\dot{\mathbf{u}}_k^{(n+1)} = \frac{\gamma}{\beta \Delta t} \mathbf{u}_k^{(n+1)} + (\text{terms known at } t_n). \quad (12)$$

Let $c_v = \gamma/(\beta \Delta t)$. The self term $Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k$ therefore produces the tangent contribution

$$\mathbf{K}_{\text{is}} = (Z c_v + \alpha) \mathbf{W}, \quad (13)$$

which is assembled into the global Hessian at each Newton iteration. In the quasi-static limit Norma.jl sets $c_v = 0$, so (13) reduces to $\alpha \mathbf{W}$ and the impedance BC collapses onto RR.

The self internal-force contribution at the current state is

$$\mathbf{f}_{\text{is}} = \mathbf{W} (Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k), \quad (14)$$

added by `build_impedance_schwarz_force` (schwarz.jl:311-338).

The partner contribution is accumulated into the boundary-force vector:

$$\mathbf{f}_{\Gamma,k} \leftarrow \mathbf{f}_{\Gamma,k} - \mathbf{t}_j^{(n)}|_{\Gamma} + Z \mathbf{W} \mathbf{P} \dot{\mathbf{u}}_j^{(n)} + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j^{(n)}, \quad (15)$$

as implemented in `apply_bc_detail` for `SolidMechanicsImpedanceSchwarzBoundaryCondition` (schwarz.jl:71-139).

3.4 Derivation of Z from material properties

In `Norma.jl` the impedance is computed per-domain from the material parameters:

$$Z = \sqrt{\rho(\lambda + 2\mu)} = \rho c_p, \quad \text{with } c_p = \sqrt{(\lambda + 2\mu)/\rho}. \quad (16)$$

No user tuning is required for the impedance term; only the optional displacement-penalty α is exposed via the `robin` parameter field.

4 Overlap Schwarz

On an overlap boundary Γ_k , the partner field $\mathbf{u}_j$ is available but the partner *traction* is not, because Γ_k is not a shared interface: it is a cut through the interior of Ω_j . Only those coupling conditions that do not require a partner traction can be imposed on an overlap boundary. `Norma.jl` provides two such families.

4.1 Classical overlap (Dirichlet coupling)

The canonical overlap Schwarz BC replaces the free boundary of Ω_k on Γ_k with an imposed kinematic trace drawn from the partner,

$$\mathbf{u}_k = \mathbf{u}_j, \quad \dot{\mathbf{u}}_k = \dot{\mathbf{u}}_j, \quad \ddot{\mathbf{u}}_k = \ddot{\mathbf{u}}_j \quad \text{on } \Gamma_k. \quad (17)$$

The trace is constructed in one of two ways:

Strong interpolation. For each overlap node $\mathbf{x}_i \in \Gamma_k$, find the partner element containing $\mathbf{x}_i$ in its reference configuration, evaluate the partner shape functions $\mathbf{N}_j(\mathbf{x}_i)$, and assign

$$\mathbf{u}_k(\mathbf{x}_i) = \sum_a \mathbf{N}_j^a(\mathbf{x}_i) \mathbf{u}_j^a.$$

The corresponding degrees of freedom on Ω_k are then frozen.

Weak (L^2) projection. A Dirichlet projector $\mathbf{P}$ is precomputed by solving the boundary L^2 problem $\mathbf{W}_k \hat{\mathbf{u}}_k = \mathbf{C}_{kj} \mathbf{u}_j$, where $\mathbf{W}_k$ is the mass matrix of Γ_k and $\mathbf{C}_{kj}$ is the cross-mass coupling the partner's shape functions to the overlap boundary. Overlap-node displacements are then assigned $\mathbf{u}_k = \mathbf{P} \mathbf{u}_j$.

Both variants are implemented by `SolidMechanicsOverlapSchwarzBoundaryCondition` and selected via the `use_weak` flag (`coupling_strong_dbc`, `coupling_weak_overlap_dbc` in `schwarz.jl`:453--506). Because (17) constrains the overlap DOFs, classical overlap adds no interface tangent term: the constrained rows and columns are eliminated from the system.

4.2 Impedance overlap

The impedance overlap BC, by contrast, leaves the overlap DOFs *free* and couples Ω_k to the partner through an impedance/Robin penalty,

$$Z (\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha (\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0} \quad \text{on } \Gamma_k. \quad (18)$$

No traction term appears on either side of (18), which is the essential discrete difference from the non-overlap impedance condition (10). On Ω_k 's side, the elastic equilibrium on Γ_k supplies the natural (free-surface) condition $\mathbf{t}_k = \mathbf{0}$; (18) then drives the self state towards the partner's state via a spring-dashpot coupling tuned to the material impedance.

The discrete assembly is the same as in the non-overlap impedance case, minus the partner traction:

- **Self tangent.** $\mathbf{K}_{\text{is}} = (Z c_v + \alpha) \mathbf{W}$, with $c_v = \gamma/(\beta \Delta t)$ under Newmark, assembled by `build_impedance_overlap_schwarz_stiffness` (schwarz.jl:265–293).
- **Self internal force.** $\mathbf{f}_{\text{is}} = \mathbf{W} (Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$, assembled by `build_impedance_schwarz_force` (schwarz.jl:311–338), shared with the non-overlap impedance BC.
- **Partner right-hand side.** $\mathbf{f}_\Gamma \leftarrow \mathbf{f}_\Gamma + \mathbf{W} (Z \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$, where $\dot{\mathbf{u}}_j$ and $\mathbf{u}_j$ are obtained by strong interpolation from the partner element containing each overlap node (`apply_bc_detail` for `SolidMechanicsImpedanceOverlapSchwarzBoundaryCondition`, schwarz.jl:216–263).

At the Schwarz fixed point $\mathbf{u}_k = \mathbf{u}_j$ and $\dot{\mathbf{u}}_k = \dot{\mathbf{u}}_j$ on Γ_k ; the self and partner impedance contributions cancel, leaving the overlap surface as a free (traction-free) surface whose interior solution equals the partner's. Because the impedance matches the material's own P-wave impedance, outgoing waves in Ω_k see no reflection at Γ_k : they exit Ω_k consistently with the solution that continues inside Ω_j .

Norma.jl additionally allows a step-dependent multiplier on Z through the `impedance_scale` vector (schwarz.jl:206–214). This is useful for ramping the impedance coupling in during the early part of a simulation to avoid exciting transients.

4.3 Why there is no Robin-Robin overlap variant

An RR condition, $\mathbf{t} + \alpha \mathbf{u} = \mathbf{g}$, requires a partner traction to build its right-hand side $\mathbf{g} = -\mathbf{t}_j + \alpha \mathbf{u}_j$. On an overlap boundary there is no partner traction to borrow. Formally dropping the $-\mathbf{t}_j$ term from RR yields

$$\mathbf{t}_k + \alpha \mathbf{u}_k = \alpha \mathbf{u}_j \quad \Longleftrightarrow \quad \mathbf{t}_k + \alpha (\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0},$$

which is the $Z = 0$ special case of (18): a pure displacement penalty with no dynamic absorption. This is subsumed by the impedance overlap BC (set $Z = 0$, keep α) and is not exposed as a separate option.

5 Comparison

5.1 Side-by-side formulation

Non-overlap variants.

	RR	Impedance
Boundary condition	$\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$	$\mathbf{t}_k + Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$
Self tangent	$\alpha \mathbf{W}$	$(Z c_v + \alpha) \mathbf{W}$
Self internal force	$\alpha \mathbf{W} \mathbf{u}_k$	$\mathbf{W} (Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$
Partner RHS	$-\mathbf{t}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$	$-\mathbf{t}_j + Z \mathbf{W} \mathbf{P} \dot{\mathbf{u}}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$
Partner traction $\mathbf{t}_j$	used	used
Z set by	(not used)	material (auto)
α set by	user	user, optional
Quasi-static limit	same	collapses to RR

Overlap variants.

	Classical (DBC)	Impedance
Boundary condition	$\mathbf{u}_k = \mathbf{u}_j$ (strong or L^2)	$Z (\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha (\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0}$
Self tangent	(DOFs eliminated)	$(Z c_v + \alpha) \mathbf{W}$
Self internal force	(none)	$\mathbf{W} (Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$
Partner RHS	$\mathbf{u}_k \leftarrow \mathbf{P} \mathbf{u}_j$	$\mathbf{W} (Z \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$
Partner traction $\mathbf{t}_j$	(not available)	(not used)
Z set by	(not used)	material (auto), optional scale
α set by	(not used)	user, optional
Quasi-static limit	same	pure α -penalty ($c_v = 0$)

5.2 Reflection at a non-overlap interface

The reflection analysis below is for the non-overlap geometry, where a single shared interface carries both the self and the partner traction. In the overlap case each subdomain's coupling boundary is embedded in the partner's interior, so the analysis below does not apply directly; Section 4 discusses the overlap behavior separately.

Place the interface at $x = 0$ and the subdomain Ω_k in $x < 0$, so the outward normal of Ω_k at Γ is $+\hat{x}$. Look for harmonic plane-wave solutions of the 1D wave equation: a right-going incident wave u_{inc} propagating out of Ω_k toward Γ , and a left-going reflected wave u_{ref} propagating back into Ω_k ,

$$u_{\text{inc}}(x, t) = F e^{i(\omega t - kx)}, \quad u_{\text{ref}}(x, t) = R e^{i(\omega t + kx)}.$$

Here ω is the angular frequency, $k = \omega/c_p$ is the wavenumber from the 1D elastic dispersion relation, and F, R are complex amplitudes. Substituting into $\sigma = M u_x$ and using $M k = M \omega/c_p = \rho c_p^2 \cdot \omega/c_p = Z \omega$ gives the interface values

$$u(0, t) = (F + R) e^{i\omega t}, \quad \sigma(0, t) = i\omega Z (R - F) e^{i\omega t}, \quad \dot{u}(0, t) = i\omega (F + R) e^{i\omega t}.$$

RR boundary condition. Imposing $\sigma + \alpha u = 0$ at $x = 0$ gives

$$i Z \omega (R - F) + \alpha (F + R) = 0,$$

and solving for the reflection coefficient $R_{\text{RR}}(\omega) \equiv R/F$ yields

$$R_{\text{RR}}(\omega) = \frac{i\omega Z - \alpha}{i\omega Z + \alpha}. \quad (19)$$

For any real $\alpha > 0$ the numerator and denominator have the same modulus, so $|R_{\text{RR}}(\omega)| \equiv 1$: the elastic interface *fully reflects* the incoming wave with a frequency-dependent phase shift. No wave energy crosses the interface.

Impedance boundary condition. Imposing $\sigma + Z \dot{u} + \alpha u = 0$ at $x = 0$ gives

$$\mathrm{i} Z \omega (R - F) + \mathrm{i} Z \omega (F + R) + \alpha (F + R) = 0,$$

which simplifies to $(2 \mathrm{i} Z \omega + \alpha) R + \alpha F = 0$, so

$$R_{\text{imp}}(\omega) = -\frac{\alpha}{2 \mathrm{i} \omega Z + \alpha}. \quad (20)$$

With $\alpha = 0$ the pure impedance condition is satisfied identically by an outgoing wave, and $R_{\text{imp}} \equiv 0$: the wave is absorbed at all frequencies. For $\alpha > 0$, $|R_{\text{imp}}(\omega)| = \alpha / \sqrt{\alpha^2 + 4 \omega^2 Z^2}$, which is strictly below one at every frequency and tends to zero as $\alpha \rightarrow 0$.

Comparison. The contrast between (19) and (20) is the key difference between the two Schwarz variants. Classical RR conserves energy at the interface ($|R_{\text{RR}}| \equiv 1$), reflecting outgoing waves back into the originating subdomain; impedance Schwarz dissipates outgoing-wave energy into the partner ($|R_{\text{imp}}| < 1$, and $R_{\text{imp}} = 0$ at $\alpha = 0$). This is the mechanism by which impedance Schwarz eliminates the reflection-driven instability that plagues classical RR in time-dependent problems.

5.3 Energy behavior with mixed integrators

With classical RR coupling across an implicit/explicit interface, reflections from the elastic interface can beat against each other and cause non-monotone energy growth over many Schwarz iterations. The impedance variant shunts the outgoing wave energy into the partner and removes that pathway, which stabilizes dynamic simulations that integrate the two subdomains differently (e.g. Newmark + central difference) without requiring a global time step.

5.4 Quasi-static limit

For quasi-static problems ($\dot{\mathbf{u}} \rightarrow \mathbf{0}$, $c_v = 0$ in (13)), the impedance condition (10) reduces exactly to the Robin condition (3). Normajl’s quasi-static branch computes $c_v = 0$ explicitly and the impedance pipeline assembles the same $\alpha \mathbf{W}$ tangent as RR. In particular, an impedance run with $\alpha = 0$ in the quasi-static limit degenerates to a pure Neumann condition and does not couple the subdomains through displacement, which is an important edge case to avoid in static use.

6 Summary

- On a *non-overlap* interface, RR Schwarz enforces an elastic condition $\mathbf{t} + \alpha \mathbf{u} = \mathbf{g}$; α is a user-supplied stiffness that trades off traction and displacement error.
- Impedance non-overlap Schwarz augments the Robin condition with the Lysmer-Kuhlemeyer absorbing term $Z \dot{\mathbf{u}}$ using the material’s own P-wave impedance $Z = \sqrt{\rho(\lambda + 2\mu)}$. No tuning is required for Z ; the displacement-penalty α is optional. Discretely, the only differences from RR are (i) an extra $Z c_v$ contribution to the interface tangent and (ii) a partner velocity term in the right-hand side; both are assembled with the same surface operator $\mathbf{W}$ and projection $\mathbf{P}$ used by RR.

- In the quasi-static limit $c_v \rightarrow 0$, impedance non-overlap Schwarz collapses onto RR. For dynamic problems it is non-reflecting at the interface ($|R_{\text{imp}}| < 1$, with $R_{\text{imp}} = 0$ at $\alpha = 0$), whereas RR is fully reflecting ($|R_{\text{RR}}| \equiv 1$). This is the mechanism that removes the reflection-driven instability of classical RR with mixed integrators.
- On an *overlap* boundary, no partner traction is available, so RR has no natural analogue. Norma.jl provides two overlap variants: classical DBC coupling, which interpolates the partner displacement (strongly or in L^2) and constrains the overlap DOFs; and impedance overlap, which leaves the overlap DOFs free and couples to the partner through the spring-dashpot penalty $Z(\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha(\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0}$.
- The impedance overlap BC uses the same self-side tangent $(Z c_v + \alpha) \mathbf{W}$ and self-side internal force $\mathbf{W}(Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$ as the non-overlap variant; the only discrete difference is that the partner right-hand side carries no $-\mathbf{t}_j$ term, because the overlap boundary is a cut through Ω_j 's interior rather than a shared interface. At convergence, outgoing waves in Ω_k exit through the overlap surface without reflection, matched to the solution that continues inside Ω_j .